

Sample Spaces and The Counting Principle

Represent the sample space using set notation.

- 1) A sandwich shop has three types of sandwiches: ham, turkey, and chicken.
- 2) The chess club must decide when to meet for a practice. The possible days are Tuesday, Wednesday, or Thursday.

 $\{\text{ham, turkey, chicken}\}$ $\{\text{Tuesday, Wednesday, Thursday}\}$

- 3) The chess club must decide when to meet for a practice. The possible days are Tuesday, Wednesday, or Thursday. The possible times are 3, 4, or 5 p.m.
- 4) When a button is pressed, a computer program outputs a random odd number greater than 1 and less than 9. You press the button twice.

 $\{(T, 3), (T, 4), (T, 5), (W, 3), (W, 4), (W, 5), (R, 3), (R, 4), (R, 5)\}$ $\{(3, 3), (3, 5), (3, 7), (5, 3), (5, 5), (5, 7), (7, 3), (7, 5), (7, 7)\}$

- 5) A spinner can land on either red or blue. You spin and then roll a six-sided die.
- 6) There are two boys and a girl on a trivia team. Two questions remain. One team member is randomly picked to answer the first question and a different member is picked to answer the second question.

 $\{(R, 1), (R, 2), (R, 3), (R, 4), (R, 5), (R, 6), (B, 1), (B, 2), (B, 3), (B, 4), (B, 5), (B, 6)\}$ $\{(B_1, B_2), (B_1, G), (B_2, B_1), (B_2, G), (G, B_1), (G, B_2)\}$ **Find the number of possible outcomes in the sample space.**

- 7) A jewelry store sells gold and platinum rings. Each ring is fitted with a ruby, sapphire, emerald, or diamond gemstone.
- 8) A spinner can land on either red, blue, or green. You spin twice.

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- 9) Eight rooms in a house need to be painted. Each room can be painted white or yellow.
- 10) Six books need to be placed on a shelf. You randomly arrange the books on the shelf from left to right.

256

720